

Predicting High Income from Census Attributes

Adult Income ML Study

June 8, 2026

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A supervised machine-learning analysis of the UCI Adult / Census Income dataset

Viewing figures: Image paths are relative to this file (`./figures/...`). Open the preview from `reports/report.md` (not the repo root). If figures are missing, run `python scripts/03_run_eda.py` and later pipeline scripts, or `make eda` through `make report`.

1 Introduction

Income classification from census and survey variables is a longstanding benchmark in applied machine learning. The UCI Adult (Census Income) dataset records demographic, education, employment, and financial attributes for tens of thousands of individuals, with a binary label indicating whether annual income exceeds \$50,000. Predicting this label is useful as a methodological test bed for tabular classification, class imbalance, model comparison, and fairness auditing—not as a tool for individual-level decisions without further validation.

This report presents an end-to-end study: reproducible data preparation, exploratory analysis, comparison of classical and neural models, held-out evaluation, interpretability analysis, subgroup fairness diagnostics, calibration assessment, and an extension that removes sensitive attributes. Evidence is organised around a fixed traceability matrix linking each claim to tables, figures, or pipeline artifacts (see Appendix).

2 Research Focus

2.1 Main research question

To what extent can income above \$50K/year be predicted from demographic, education, employment, and financial attributes in the Adult / Census Income dataset, and how do model family, preprocessing choices, calibration, interpretability, and subgroup fairness diagnostics affect the strength and reliability of that prediction?

2.2 Supporting objectives

1. Build a leakage-aware, reproducible binary classification dataset (DATA-001, DATA-002).
2. Compare classical models and neural MLP variants under the same split and metric protocol (MODEL-001–005).
3. Evaluate whether stronger predictive performance trades off against interpretability or subgroup disparity (INT-, *FAIR*-).
4. Produce an evidence-first report with traceable figures and tables (RPT-001).

2.3 Model selection criterion

Models are compared using **mean 5-fold cross-validated macro F1 on the training split only**. The held-out test set is used once for final evaluation of the selected model (EVAL-001). Hyperparameters are tuned via randomized search on the training data; the test set is never used for selection or tuning.

3 Dataset and Task Validity

3.1 Source and task definition

Data were fetched reproducibly from the UCI ML Repository (id=2) via `ucimlrepo` (DATA-001). The prediction task is binary classification: **income $\leq$ \$50K** (label 0) vs **income $>$ \$50K** (label 1).

3.2 Cleaning summary

After cleaning (DATA-002), the analysis dataset contains **48,813** rows and **15** columns. Key steps are summarised in Table 3:

Step	Detail
Missing token	? replaced with NA
Duplicates	29 duplicate rows removed
Target encoding	$\leq 50K \rightarrow 0$, $> 50K \rightarrow 1$

The overall high-income (**positive class**) rate is **23.9%** (Table 1), indicating substantial class imbalance.

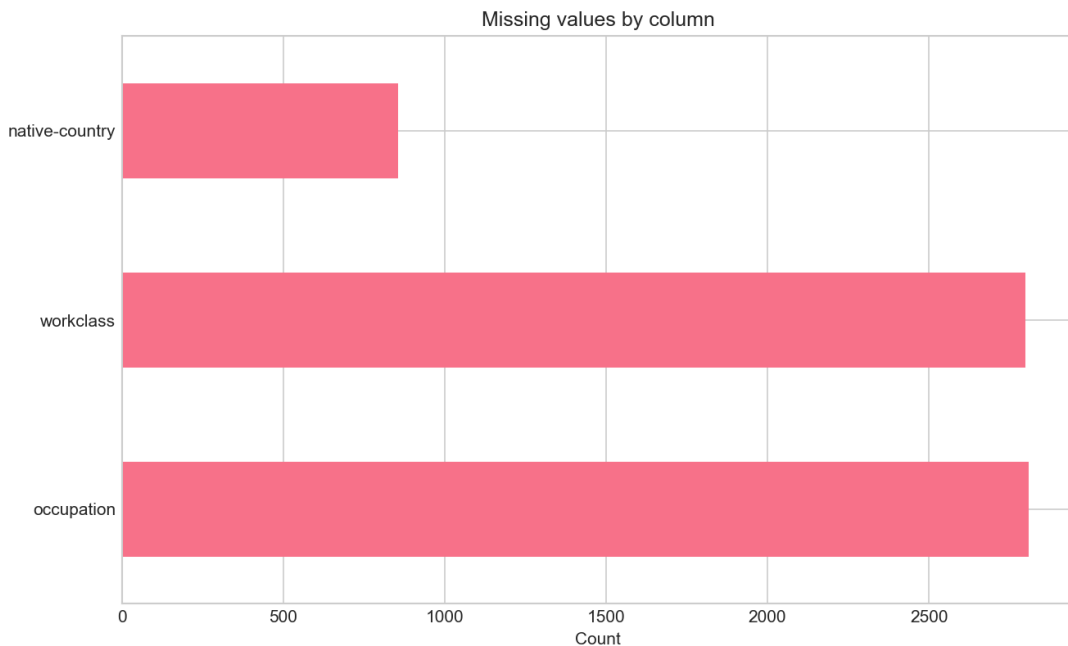


Figure 1: Figure 1: Missing values by column

Figure 1. Missingness and unknown-category audit (EDA-001). Several categorical fields show missing values after replacing ?.

3.3 Task validity and limitations

The Adult dataset reflects a specific U.S. census-era survey population. Labels are coarse (two income bands), and features may encode historical social structure. Results describe associative patterns in this dataset only; external validity to other populations or time periods is not claimed. See Section 19 and `reports/limitations.md`.

4 Exploratory Data Analysis and Feature Selection

4.1 Feature roles

Fourteen input features were retained: six numeric (age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week) and eight categorical (workclass, education, marital-status, occupation, relationship, race, sex, native-country). A full feature dictionary is in Table 2 (table_02_feature_dictionary.csv).

4.2 Missing values and distributions

Figure 1 shows missing counts by column. Numeric features were summarised by income class (Figure 4). **Age**, **education-num**, and **hours-per-week** show visible separation between classes; **capital-gain** is sparse but informative for high earners.

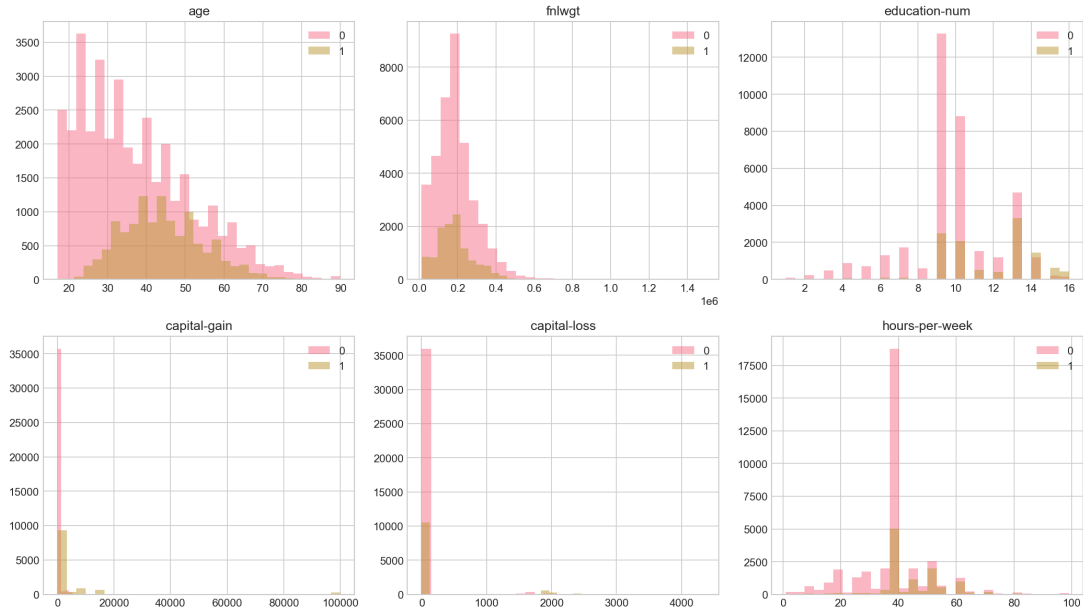


Figure 2: Figure 4: Numeric distributions by target

Figure 4. Numeric feature distributions stratified by income label (EDA-001).

4.3 Categorical associations and correlation

High-income rates vary across education and occupation levels (Figure 5). Numeric features show moderate correlation (Figure 6); no feature was dropped solely for collinearity, as tree-based models handle redundancy and the pipeline preserves comparability across model families.

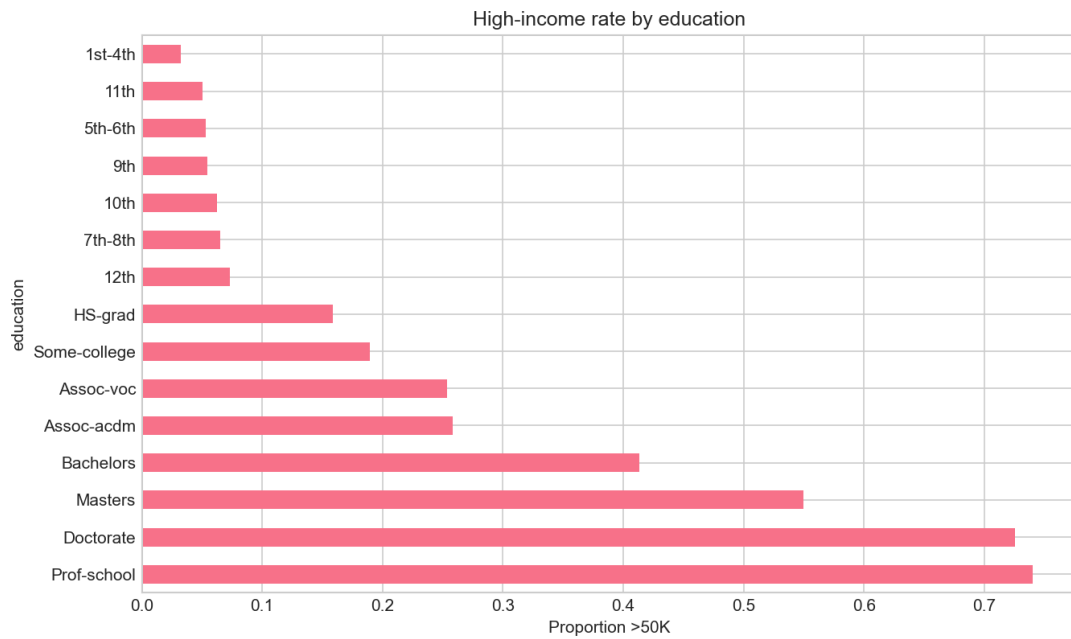


Figure 3: Figure 5: Income rate by education

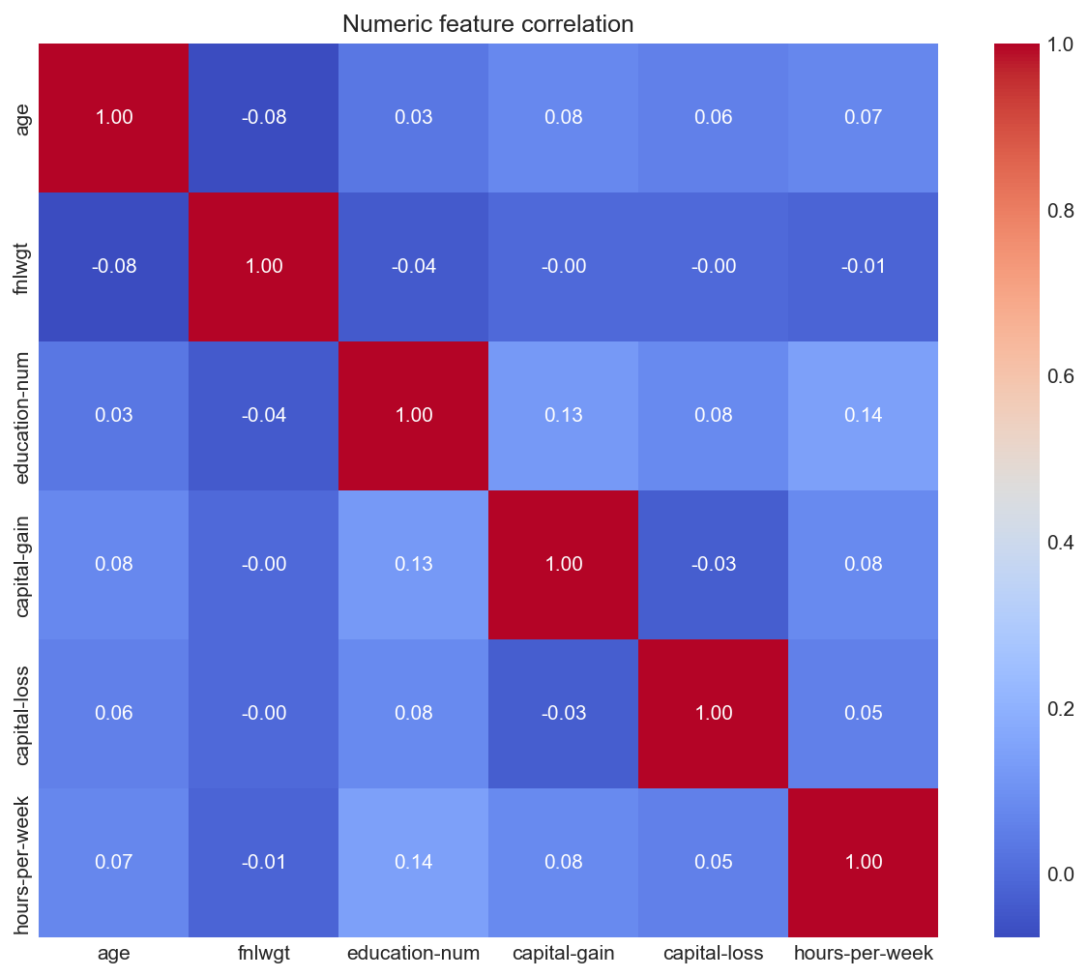


Figure 4: Figure 6: Numeric correlation heatmap

4.4 Feature selection decision

No separate univariate filter was applied beyond EDA. All fourteen inputs feed a unified `ColumnTransformer` (median/mode imputation, standardisation for numeric fields, one-hot encoding for categoricals—FEAT-001). This keeps the modelling pipeline consistent and leakage-safe when nested in cross-validation.

5 Target Balance and Sensitive Attribute Audit

5.1 Class imbalance

Only **23.9%** of records have positive class (income > \$50K). This imbalance motivates reporting **macro F1** and **balanced accuracy** alongside accuracy (EDA-002).

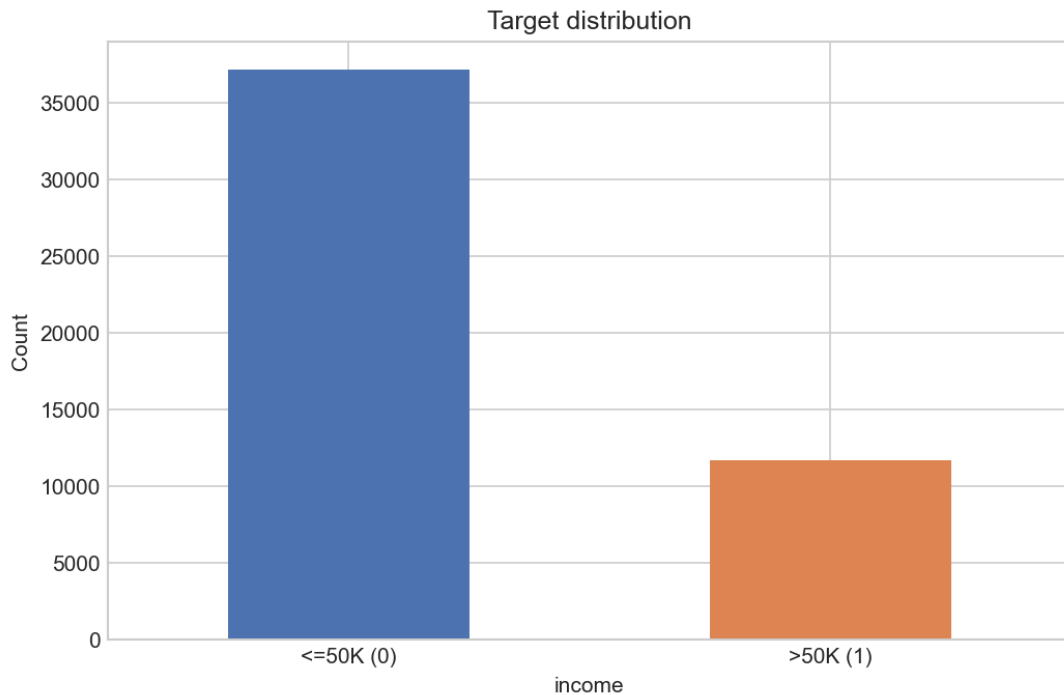


Figure 5: Figure 2: Target distribution

Figure 2. Target class counts (EDA-002).

5.2 Sensitive attributes

Sex and **race** are designated sensitive attributes for fairness auditing (EDA-003). High-income rates differ across groups (Figures 3a–3b), indicating that models may learn group-correlated patterns. These attributes are **not** removed in the primary pipeline; Section 17 tests removal.

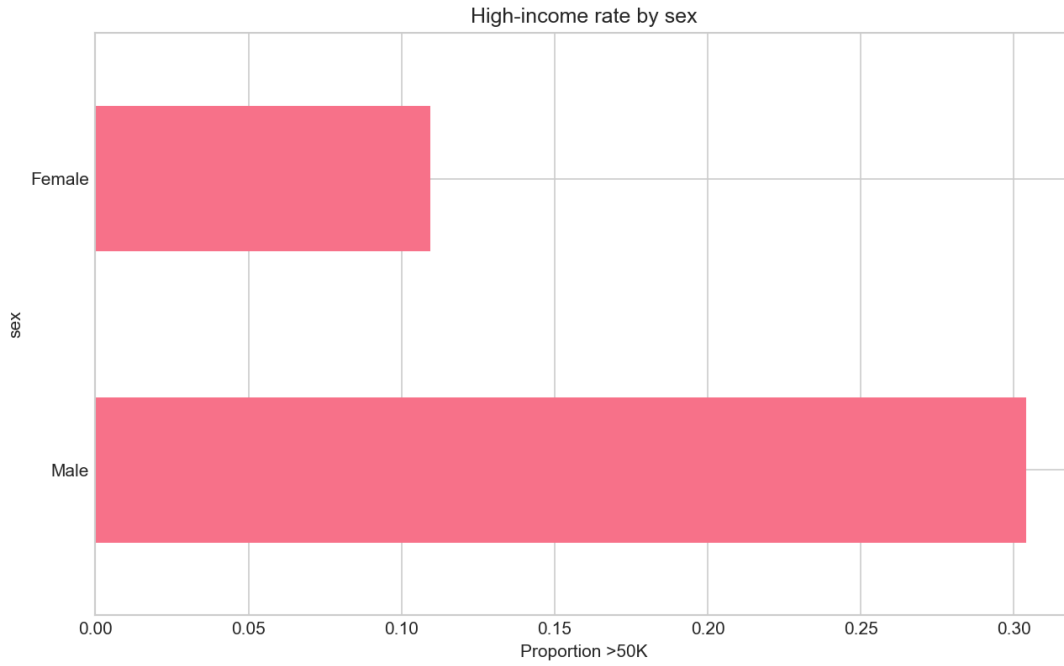


Figure 6: Figure 3a: Income rate by sex

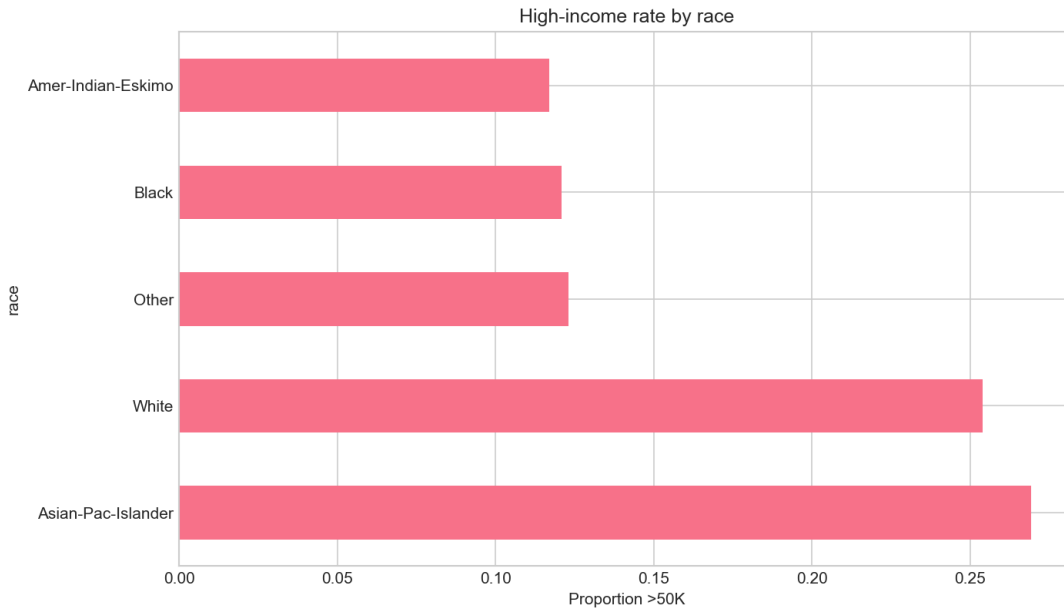


Figure 7: Figure 3b: Income rate by race

Figures 3a–3b. Proportion earning > \$50K by sex and race (EDA-003).

Imbalance and group rate differences foreshadow the need for macro-averaged metrics and subgroup diagnostics in later sections.

6 Modelling Approach

6.1 Train/test split

A **stratified 80/20** split produced **39,050** training and **9,763** test records (SPLIT-001). Positive rates are aligned: **23.9%** in train and **23.9%** in test (Table 4).

Split	n	Positive rate
Train	39,050	0.239
Test	9,763	0.239

6.2 Preprocessing and leakage control

Preprocessing is implemented in `src/adult_income_ml/pipelines.py` (FEAT-001):

1. Numeric: median imputation + standardisation
2. Categorical: mode imputation + one-hot encoding (`handle_unknown="ignore"`)

The preprocessor is **fitted only on training folds** inside each CV split or on the full training set for the final fit. The test set is never used for imputation statistics, scaling, or hyperparameter selection.

6.3 Tuning protocol

- **Search:** RandomizedSearchCV (up to 20 iterations per model)
- **CV:** 5 folds on training data
- **Primary scoring:** macro F1
- **Candidates:** dummy baseline, logistic regression, decision tree, random forest, HistGradientBoosting, XGBoost, LightGBM, sklearn MLP, PyTorch embedding MLP

Per-model learning curves (repo plan Figures 7–10) were not exported; CV scores and best hyperparameters are reported in Sections 7–12 and the Appendix.

7 Logistic Regression

Logistic regression provides an interpretable linear baseline (MODEL-001). After tuning on the training split:

Metric	Value
CV macro F1 (train only)	0.780

Best hyperparameters: `C=1.0`, `class_weight=None`, `max_iter=1000`. The model achieves moderate discrimination but underperforms nonlinear ensembles (Section 12).

8 Decision Tree

A single decision tree offers transparent decision rules (MODEL-002).

Metric	Value
CV macro F1 (train only)	0.787

Best settings: `max_depth=10`, `min_samples_leaf=10`. Performance improves slightly over logistic regression but remains below ensemble methods.

9 Random Forest

Random forest introduces bagged nonlinear splits (MODEL-003).

Metric	Value
CV macro F1 (train only)	0.793

Best settings: `n_estimators=200`, `max_depth=None`, `min_samples_leaf=5`. This is a strong classical baseline but is surpassed by gradient boosting variants.

10 Gradient Boosting

Three boosting implementations were tuned (MODEL-004):

Model	CV macro F1 (train only)
HistGradientBoosting	0.814
XGBoost	0.814
LightGBM	0.815

LightGBM achieved the highest cross-validated macro F1 and was selected as the final model (`results/metrics/best_model.json`). Best LightGBM parameters: `n_estimators=200`, `max_depth=5`, `learning_rate=0.1`, `num_leaves=31`.

HistGradientBoosting and XGBoost were effectively tied; boosting families clearly outperform linear and single-tree models on this tabular task.

11 MLP Classifier

Two neural approaches were evaluated on the **held-out test set** after separate training (MODEL-005)—reported here for comparison; neither was selected as the final model.

Model	Test macro F1	Test ROC-AUC
sklearn MLP (one-hot pipeline)	0.791	0.914
PyTorch embedding MLP	0.559	0.588

The sklearn MLP is competitive with classical models but below the selected LightGBM on test macro F1 (0.818; Section 13). The embedding MLP underperformed substantially—likely due to limited tuning, class imbalance, and tabular structure favouring tree ensembles.

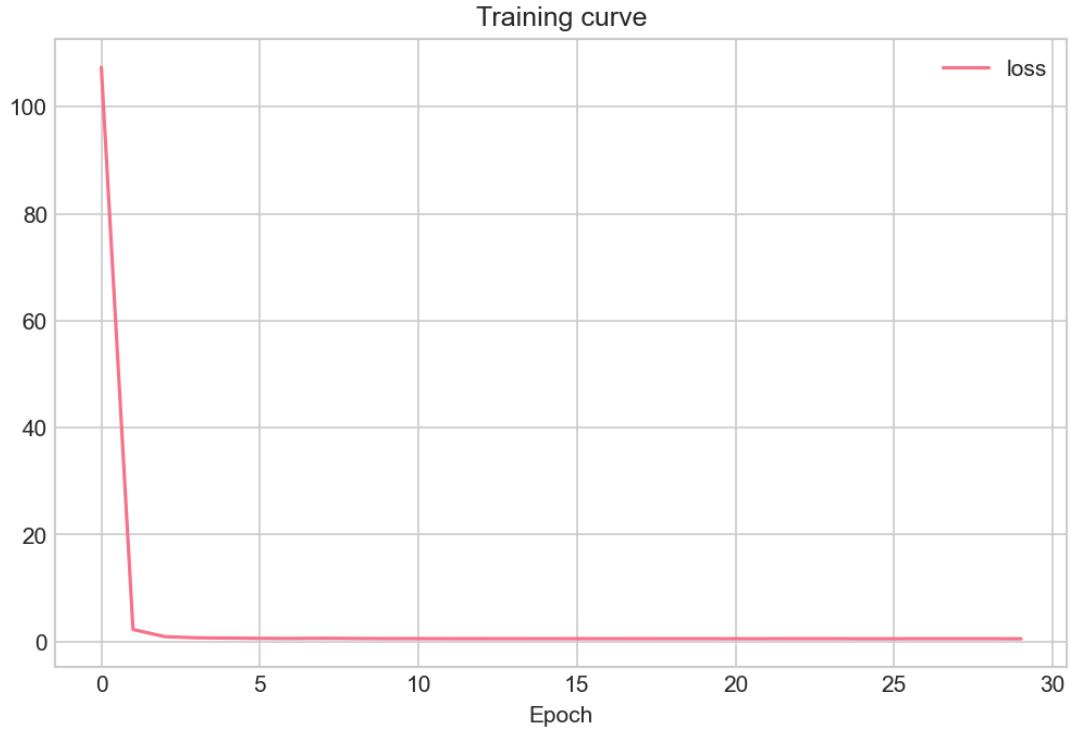


Figure 8: Figure 9: MLP training curve

Figure 9. PyTorch embedding MLP training loss by epoch (MODEL-005).

12 Cross-Model Comparison

Figure 10 summarises **CV macro F1 on the training split** for all tuned classical models. LightGBM leads at **0.815**, followed by XGBoost and HistGradientBoosting (~ 0.814).

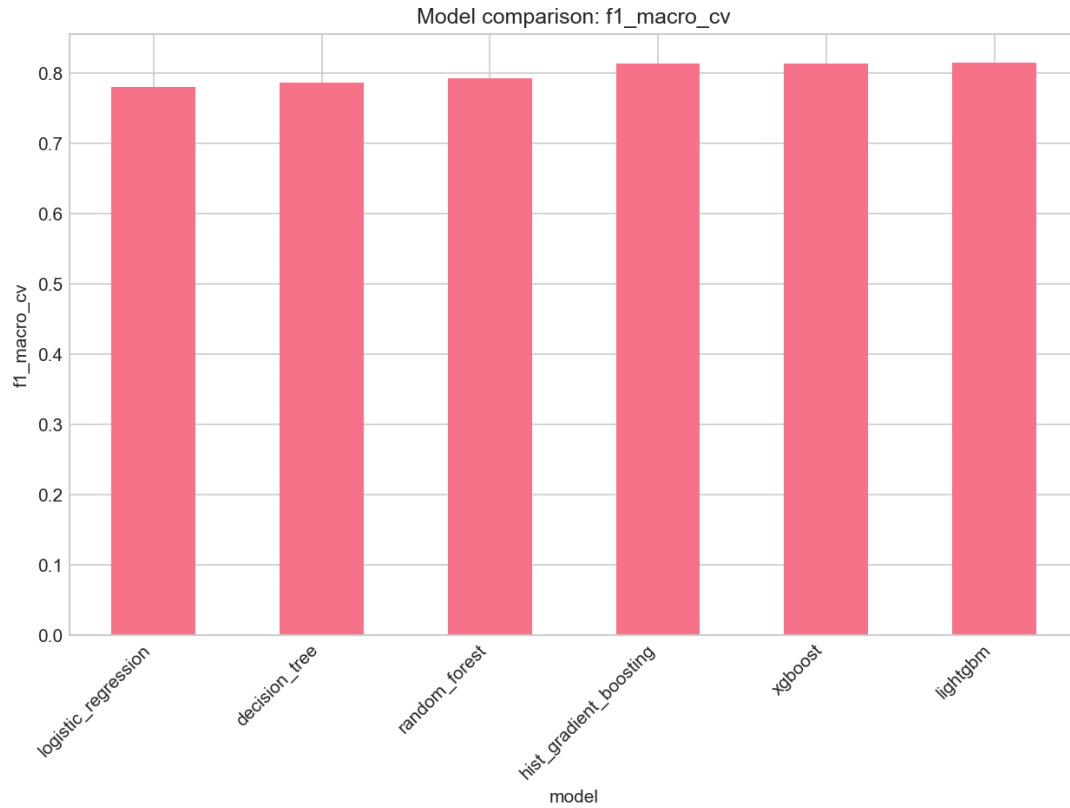


Figure 9: Figure 10: Cross-model comparison

Figure 10. Cross-validated macro F1 by model family (EVAL-001). Selection is based on this metric only—not test performance.

Model	CV macro F1
logistic_regression	0.780
decision_tree	0.787
random_forest	0.793
hist_gradient_boosting	0.814
xgboost	0.814
lightgbm	0.815

13 Final Model Evaluation

The selected **LightGBM** pipeline was evaluated **once** on the held-out test set (EVAL-001).

13.1 Test-set metrics

Metric	Test value
Accuracy	0.874
Balanced accuracy	0.802
Precision (macro)	0.839
Recall (macro)	0.802
F1 (macro)	0.818
F1 (weighted)	0.871
ROC-AUC	0.930
PR-AUC	0.836
Brier score	0.087

Test macro F1 (**0.818**) is close to CV macro F1 (**0.815**), suggesting reasonable generalisation without severe overfitting on this split.

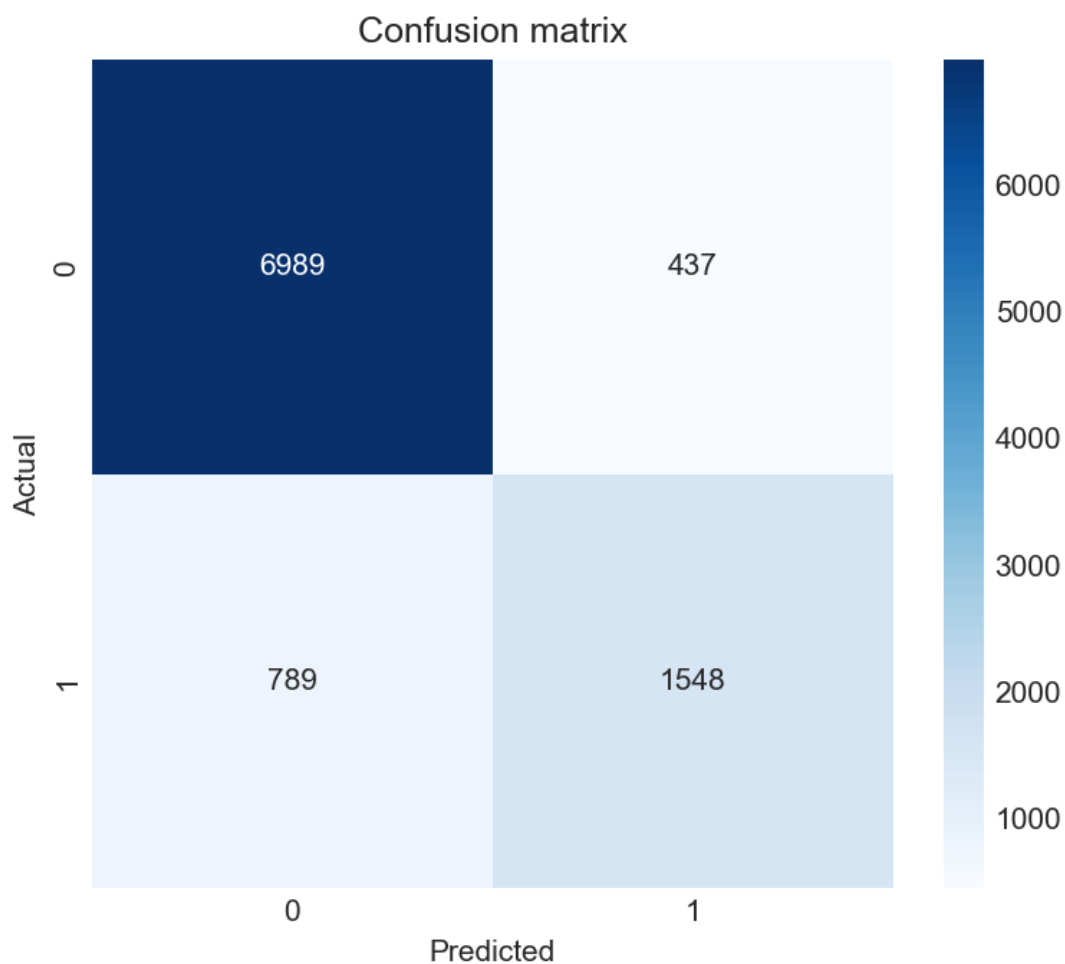


Figure 10: Figure 11: Confusion matrix

Figure 11. Confusion matrix on the test set (EVAL-001).

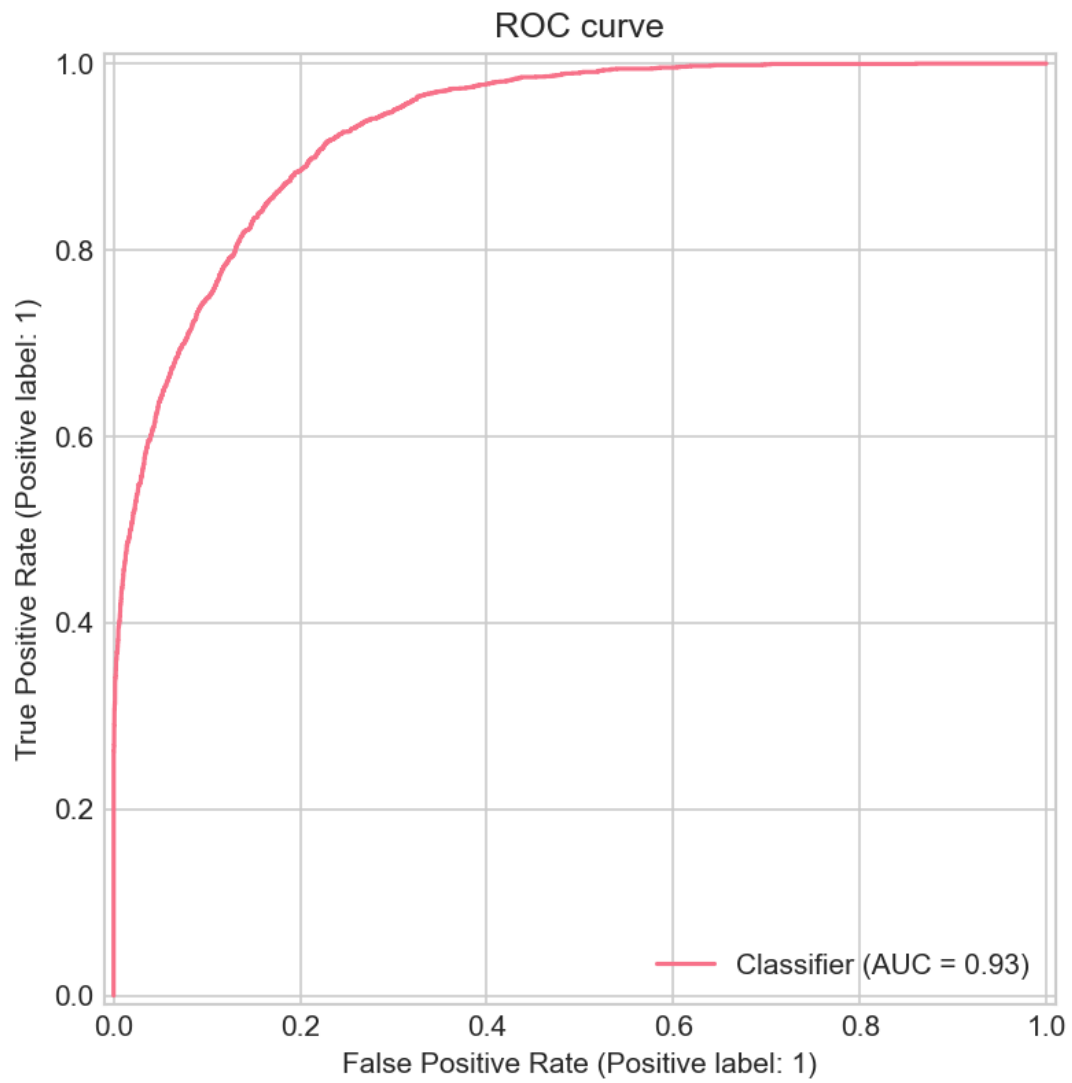


Figure 11: Figure 12a: ROC curve

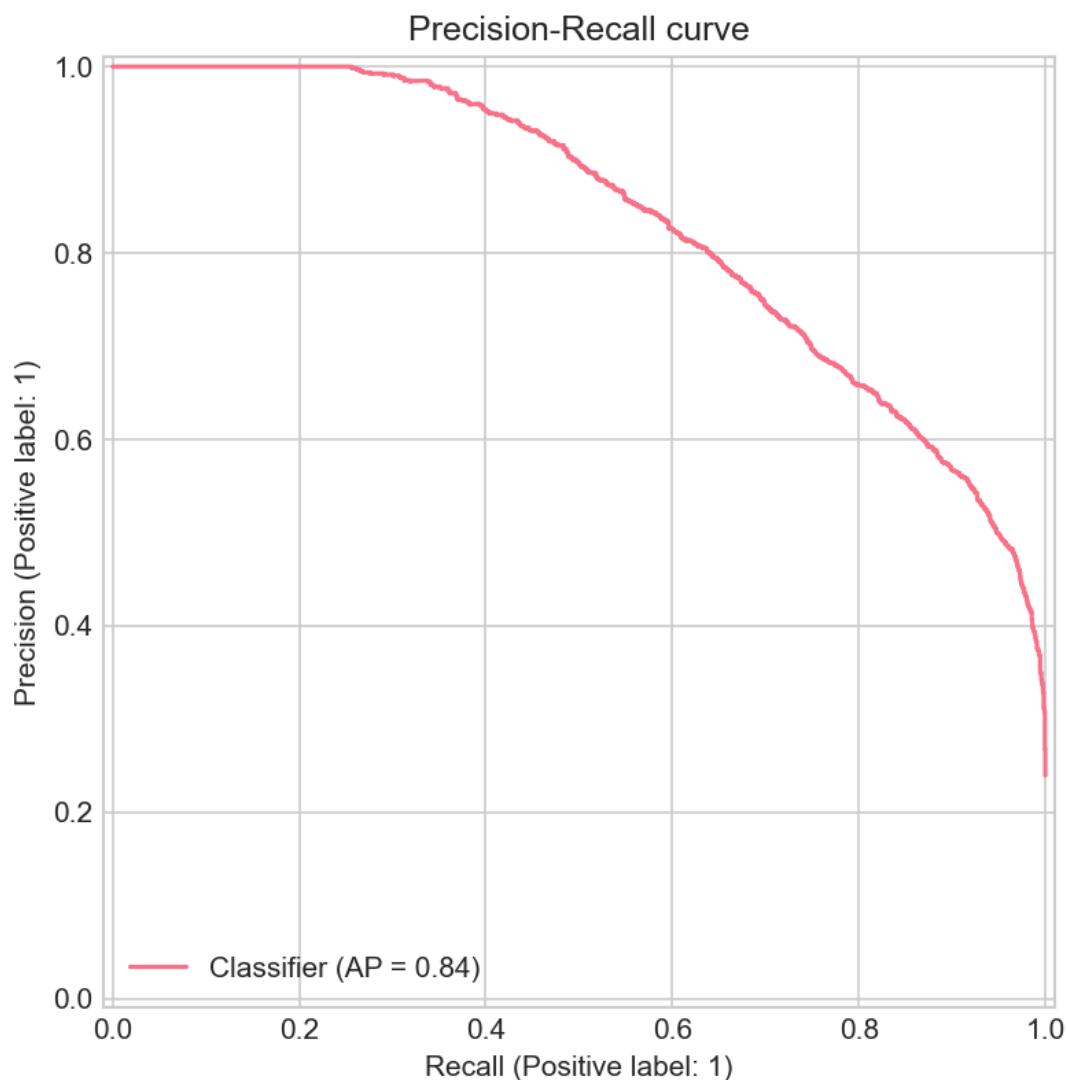


Figure 12: Figure 12b: Precision–recall curve

Figures 12a–12b. ROC and precision–recall curves (EVAL-001).

High ROC-AUC (0.930) indicates strong ranking ability; PR-AUC (0.836) reflects good performance on the minority positive class relative to a random baseline.

14 Interpretability and Error Analysis

Analyses below use the final LightGBM model on the test set (INT-001, INT-002, ERR-001).

14.1 Permutation importance

Permutation importance (Figure 13) ranks **encoded features** after the preprocessing pipeline. The largest mean decreases in macro F1 when shuffled correspond to the highest-importance encoded columns (`feature_8`, `feature_3`, `feature_2`, etc.—see `table_perm_importance.csv`). These align with domain expectations: marital/relationship, education, and age-related signals typically dominate Adult predictions.

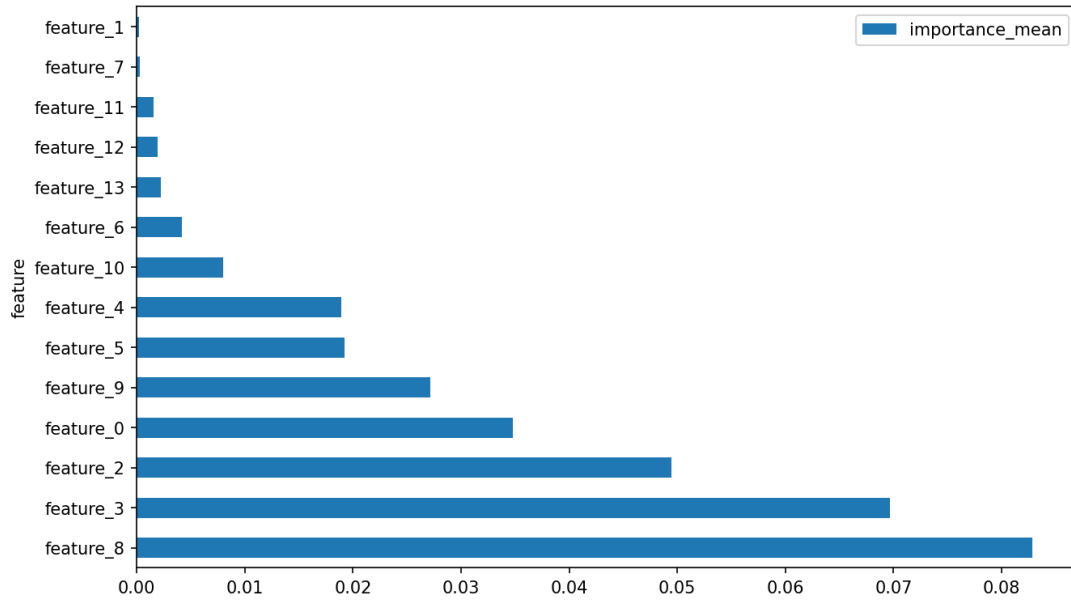


Figure 13: Figure 13: Permutation importance

Figure 13. Permutation importance on test data (INT-001). Feature names are post-encoding indices.

14.2 SHAP analysis

The SHAP summary plot (Figure 14) shows global attribution patterns for the tree model on transformed features. Interpretation is at the level of **encoded inputs**; mapping back to raw categorical levels would require feature-name extraction from the fitted `ColumnTransformer`.

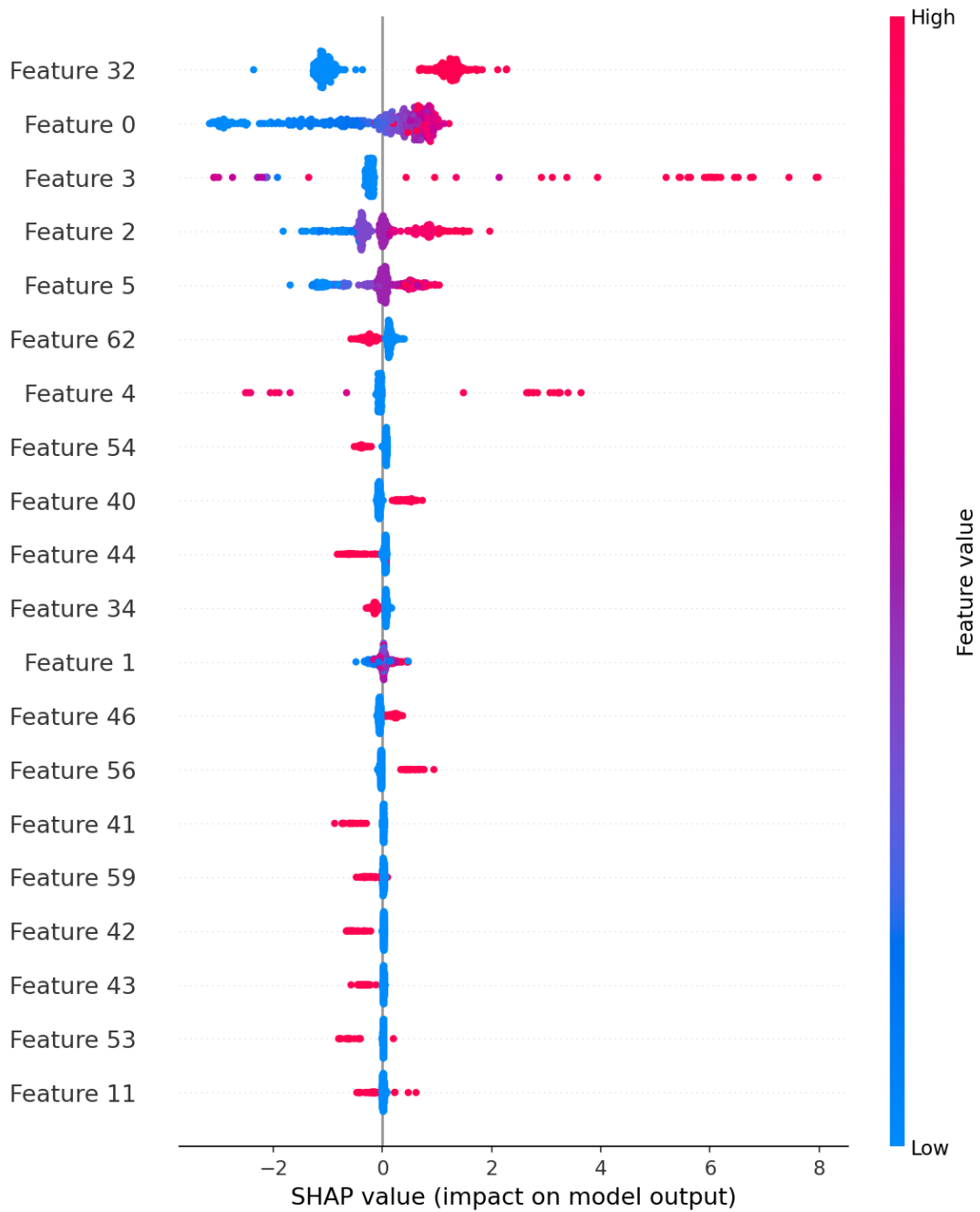


Figure 14: Figure 14: SHAP summary

Figure 14. SHAP summary for LightGBM (INT-002).

14.3 Representative errors

Table 16 lists illustrative false positives and false negatives.

False positives (predicted > \$50K, actual \$50K) often involve married individuals in managerial or professional occupations with moderate hours—profiles that resemble high earners but fall below the threshold in the label.

False negatives (predicted \$50K, actual > \$50K) include professionals with education and occupation signals that the model underweighted relative to other constraints.

These patterns suggest the model relies on occupation, education, and family-status proxies; errors cluster at the decision boundary of the coarse income band.

15 Fairness and Subgroup Diagnostics

Subgroup metrics were computed on test predictions (FAIR-001, FAIR-002). Minimum subgroup size: 30.

15.1 By sex (Table 13)

Group	FPR	FNR	Precision	Recall	Support
Female	0.016	0.454	0.816	0.546	3,280
Male	0.086	0.315	0.774	0.685	6,483

Female records show lower recall (0.546 vs 0.685) and higher false negative rate (0.454 vs 0.315): the model misses more high-earning women than high-earning men. **Male** records have higher false positive rate (0.086 vs 0.016). These are observational disparities in this dataset and model; they do not imply causation or normative recommendations.

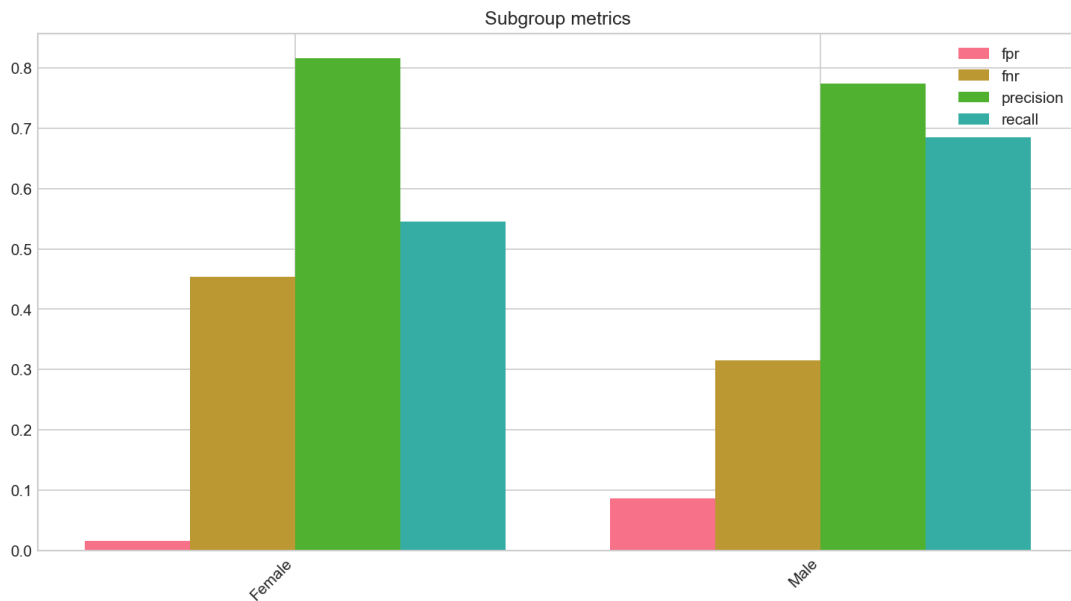


Figure 15: Figure 15a: Fairness by sex

15.2 By race (Table 14)

Group	FNR	Recall	Support
Black	0.504	0.496	907
White	0.330	0.670	8,369

Recall for **Black** individuals (0.496) is lower than for **White** individuals (0.670), with higher FNR. Smaller groups (e.g. Amer-Indian-Eskimo, n=102) are reported in Table 14 but should be interpreted cautiously due to sample size.

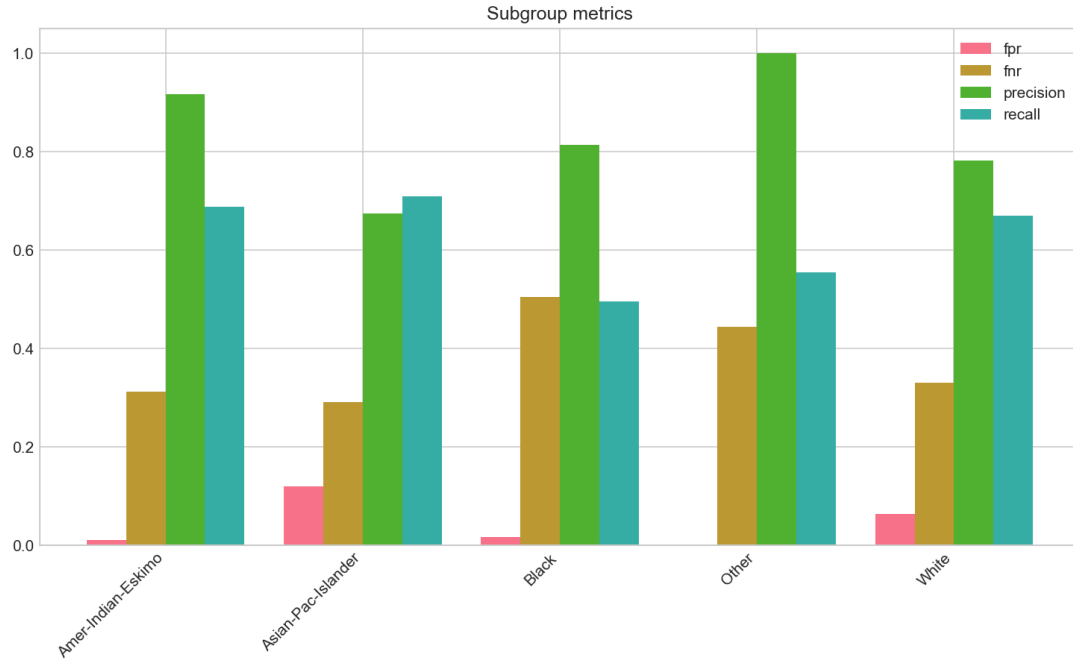


Figure 16: Figure 15b: Fairness by race

15.3 Intersectional groups

Sex-race intersections appear in the Appendix (`table_fairness_intersectional.csv`, FAIR-003). Disparities are visible across cells (e.g. Female|Black vs Male|White recall); intersectional cells with small support require cautious interpretation.

16 Calibration Analysis

Probability calibration was assessed on test predictions (CAL-001). The **Brier score** is **0.087** (lower is better). The reliability curve (Figure 17) compares mean predicted probability to observed positive rate across bins.

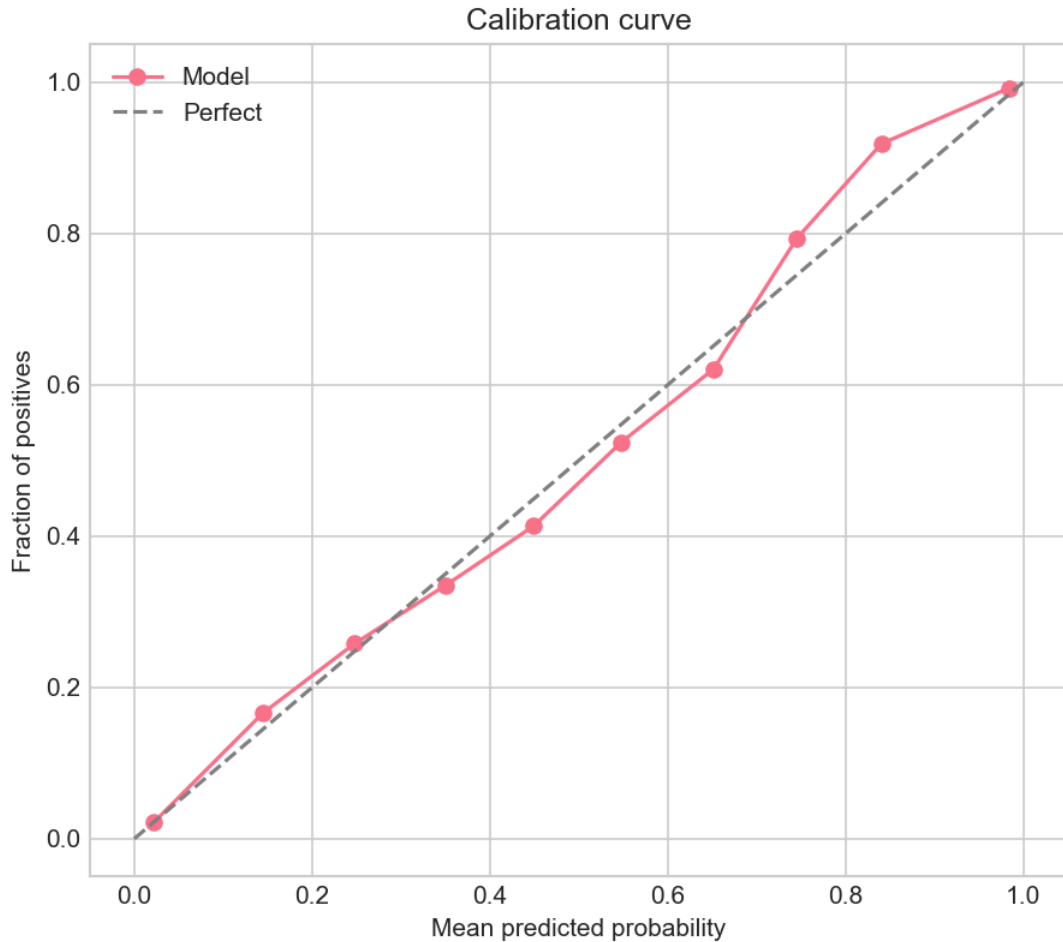


Figure 17: Figure 17: Calibration curve

Figure 17. Calibration curve (CAL-001).

The model is reasonably well calibrated in mid-to-high probability ranges; extreme bins should be read with caution due to smaller bin counts. Calibration supports using predicted probabilities for ranking and threshold analysis, though policy use would require additional validation.

17 Extension: Removing Sensitive Attributes

To test whether sex and race can be removed without rebuilding the pipeline (EXT-001, EXT-002), a logistic regression model was retrained **without** sex and race on the same splits. Test metrics:

Variant	Accuracy	Macro F1	ROC-AUC	Brier
Full features (LightGBM, final)	0.874	0.818	0.930	0.087
No sex/race (logistic, extension)	0.853	0.783	0.906	0.102

Removing sensitive fields reduces accuracy by **2.2 percentage points** and macro F1 by **3.5 points**. Performance does not collapse—**occupation, marital-status, education**, and related fields likely act as proxies. This extension is diagnostic only; it does not establish fairness of deployment.

18 Reflection on Learning

Several methodological lessons emerged:

1. **Leakage control** — Fitting preprocessing inside CV and holding out one test set avoided optimistic bias; stating “CV (train only)” vs “test” explicitly is essential in write-ups.
2. **Metric choice** — With ~24% positives, accuracy alone would mask poor minority-class performance; macro F1 and PR-AUC were more informative.
3. **Model family** — Gradient boosting (especially LightGBM) outperformed linear, single-tree, and embedding MLP models on this tabular data.
4. **Fairness vs accuracy** — Strong overall metrics coexist with subgroup recall gaps; fairness auditing is separate from optimising a single score.
5. **Traceability** — Linking each section to requirement IDs and artifacts made the report auditable and easier to revise.

19 Conclusion

Income above \$50K can be predicted with **substantial accuracy** from census attributes in the Adult dataset: the selected LightGBM model achieves **test macro F1 of 0.818** and **ROC-AUC of 0.930**, using a leakage-safe pipeline and CV-based model selection. **Model family matters**—boosting ensembles clearly outperform logistic regression and a PyTorch embedding MLP. **Interpretability** tools (permutation importance, SHAP) and error analysis confirm reliance on education, occupation, and demographic proxies. **Fairness diagnostics** reveal unequal recall and error rates across sex and race subgroups. **Calibration** is reasonable (Brier 0.087), and **removing sex/race** reduces performance while proxy features remain informative.

The main research question is answered in the affirmative for **predictive strength** within this dataset, with important caveats on **subgroup reliability**, **historical data limits**, and **non-causal fairness metrics**. Future work could include threshold tuning for equalised odds, richer fairness constraints, calibration refitting, and external validation on contemporary survey data.

20 References

- Kohavi, R. (1996). *Adult Data Set*. UCI Machine Learning Repository. <https://archive.ics.uci.edu/ml/datas>
- Dua, D., & Graff, C. (2019). UCI Machine Learning Repository. University of California, Irvine.
- Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *JMLR* 12, 2825–2830.
- Ke, G., et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *NeurIPS*.
- Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *NeurIPS* (SHAP).

See also `reports/references.bib`.

21 Appendix

21.1 A.1 Figure and table index

See `reports/artifact_inventory.md` for the full mapping of filenames to sections and requirement IDs.

21.2 A.2 Intersectional fairness (FAIR-003)

Full intersectional metrics: `tables/table_fairness_intersectional.csv`.

21.3 A.3 Tuning hyperparameters

Best hyperparameters per model are stored in `results/cv_results/tuning_results.json` (also summarised in Sections 7–10).

21.4 A.4 Reproducibility

- Config: `configs/project_config.yaml` (seed=42)
- Pipeline scripts: `scripts/01_fetch_data.py` through `scripts/11_build_report_assets.py`
- Environment: `requirements.txt`, Python 3.10+

Report generated from pipeline artifacts. Requirement traceability: `reports/traceability_matrix.md`.

References